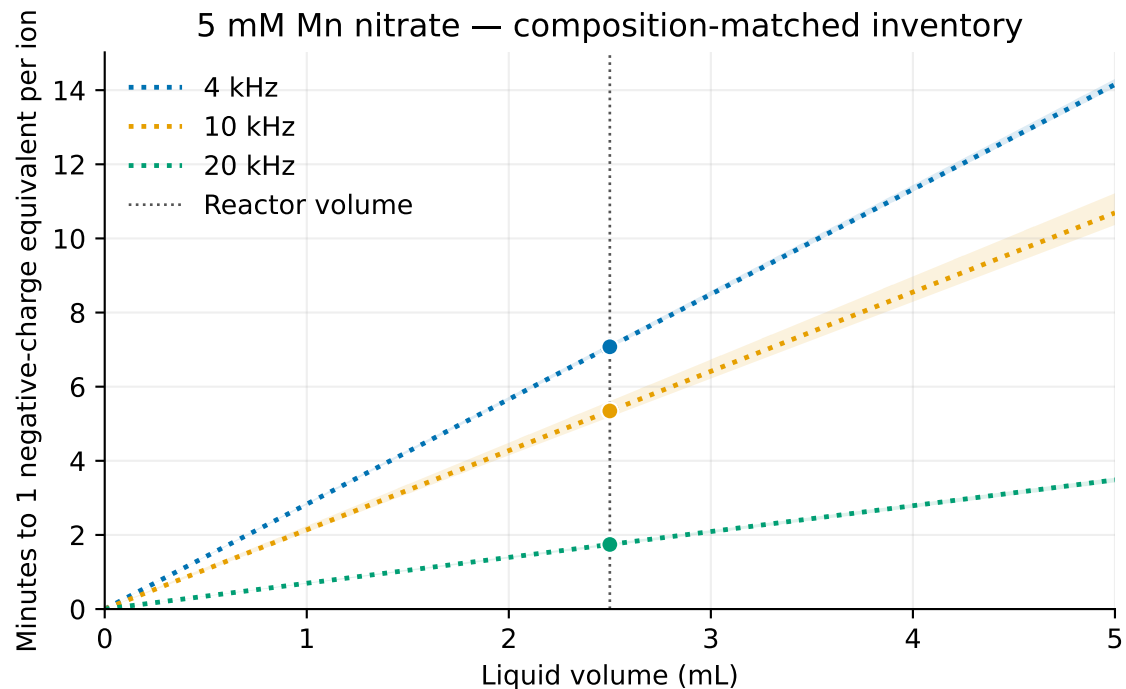
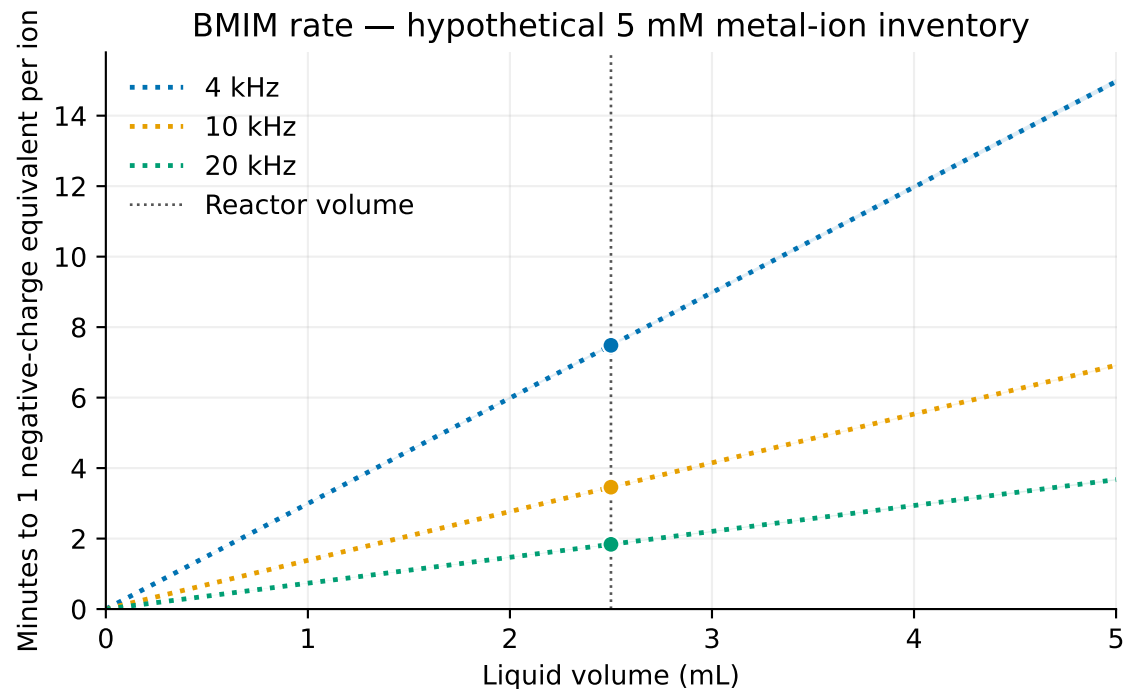


Electrical dose clock: 5 mM inventory, 1 e⁻-equivalent per ion
Curves use measured total negative charge-equivalent rates



Assumes 100% charge delivery and utilization; negative charge includes electrons and negative ions. Mn²⁺ full reduction requires 2 equivalents per ion. Dotted curves use non-reportable diagnostic rates.